

LONDON METROPOLITAN UNIVERSITY**PROFESSIONAL WORK PLACEMENT****LEARNING LOG**

YOUR ID: 22072194

YOUR NAME: Rohan Khadka

YOUR COURSE: BSc (Hons) computing

YOUR WRL TUTOR: Nishesh Biswas

PLACE OF WORK (Company Name): Lunar I.T. Solutions Pvt. Ltd.

DATES OF ENTRIES IN LEARNING LOG:

Log Index	Start Date	End Date	Company Supervisor Signature
Log 1	25 th June, 2024	4 th July, 2024	
Log2	5 th July, 2024	14 th July, 2024	
Log3	15 th July,2024	23 rd July, 2024	
Log 4	24 th July, 2024	1 st August, 2024	
Log 5	5 th August,2024	13 th August, 2024	

Log 6	15 th August, 2024	25 th August, 2024	

LEARNING LOG 1

For the period 25th June, 2024 to 4th July, 2024.

What have I done? (relate tasks to Learning Outcomes)

(LO1) – Communication Skills

During my first week at the office, I made an effort to get to know my colleagues and communicate with them. Whenever I was confused or had issues with my tasks, I asked my supervisor for help. This approach not only helped me resolve my issues efficiently but also fostered strong working relationships within my team. These interactions significantly enhanced my communication skills, making me more comfortable and effective in a professional setting.

(LO2) – Time Management Skills

During my time at the office, I was assigned various tasks that needed to be completed by the end of each day. I successfully completed all tasks on time, demonstrating effective time management skills. Additionally, I attended the office punctually every day. This punctuality and discipline ensured that I was always present and prepared to start work on time, further reinforcing my time management capabilities.

(LO5) – React.js

Throughout the week, I developed a simple website using React.js. I learned essential concepts such as props and routing and began exploring basic JavaScript integration within React components, which are crucial aspects of React. This hands-on experience allowed me to apply these concepts effectively in my project, thereby enhancing my proficiency in React.js.

(LO4) – Research skills

I conducted individual research to gather relevant information and insights for my projects. This involved finding and reviewing websites and videos related to my individual projects. I specifically researched the use of props in React.js and the basics of

JavaScript. This research helped me understand and use these concepts better in my work. It showed that I can do independent research and apply what I learn to real projects.

What I did well (refer to skills used)

Learning New Technologies

I quickly learned to use React.js, which I hadn't used much before. I spent time understanding and applying new concepts like props and routing, allowing me to develop a simple website and use basic JavaScript. This showed my ability to learn and use new technologies effectively.

Individual Research

I successfully did my own research to find information for my projects. This included looking at various websites and videos to understand new concepts. My research skills helped me solve problems on my own and improve my knowledge of React.js and JavaScript. This self-directed learning approach helped me overcome challenges and improve my technical skills.

What I could improve on (i.e. skills I want to improve)

Advanced React.js Features

While I learned the basics of React.js, I want to get better at advanced features. This includes using hooks like `useEffect` and `useContext`, and implementing custom hooks for reusable logic to make performance better. Improving these skills will help me build more complex and efficient applications.

Communication Skills

Although I managed my tasks individually, I realized that improving my communication skills could enhance the overall project outcomes. I want to get better at actively

participating in team discussions, sharing ideas, and asking for feedback. This will help me communicate more effectively with my team and supervisors.

Action I can take to improve my skills and learning (make these “SMART”):

Communication Skills

Goal: To improve my communication skills and be more confident in discussing topics I am confused about with my supervisor.

Action: Get to know my colleagues individually to feel more comfortable asking for help whenever needed. Additionally, I will confidently ask my supervisor about any topics I am confused about.

Timing: By the end of my first week at the office.

Advanced React.js Features

Goal: To enhance my understanding and application of advanced React.js features.

Action: Conduct research on advanced React.js topics through various websites, videos, and articles. I will also discuss these topics with my supervisor to deepen my understanding.

Timing: By the end of the third week of the internship.

LEARNING LOG 2

For the period 5th July, 2024 to 14th July, 2024.

What have I done? (relate tasks to Learning Outcomes)

(LO1) – Communication Skills

This week, I improved my communication by asking questions when I was confused and talking to my colleagues. I gave updates on my work and asked for feedback. This helped me clear up any confusion and work better. Talking regularly with my team also made our relationships stronger and made me feel more comfortable speaking up.

(LO5) – React.js and Tailwind CSS

I spent time learning and using Tailwind CSS to make my web pages look nice and work well on different devices. I used Tailwind CSS in my React project to style components, which made my application look better and easier to use. This practice helped me get better at designing user interfaces quickly and effectively.

(LO6) – Node.js

This week, I started learning the basics of Node.js. I set up a simple Node.js server and learned how to create basic server-side applications. I focused on key ideas like handling requests and responses and working with different modules. I also did research to build a basic API with Node.js. Learning these basics is important for building full-stack applications.

(LO4) – Research Skills

I improved my research skills by finding information online about Node.js and Tailwind CSS. I watched videos, and looked at documentation to learn the best ways to use these

tools in my projects. Doing this research helped me understand how to apply what I learned to real projects.

What I did well (refer to skills used)

Learning New Technologies

I quickly learned the basics of Node.js and Tailwind CSS, demonstrating my ability to learn new technologies efficiently. Setting up a Node.js server allowed me to understand fundamental concepts like handling requests and responses, and working with modules. This foundation is crucial for building more complex server-side applications in the future.

Applying Tailwind CSS

Using Tailwind CSS, I transformed the design of my React project into a sleek and responsive user interface. I successfully applied utility classes to create consistent and modern components, enhancing the overall user experience.

Individual Research

I successfully conducted research on Node.js and Tailwind CSS, using various online resources to gather information and understand best practices. This self-directed learning helped me solve problems and improve my technical skills.

What I could improve on (i.e. skills I want to improve)

Advanced Node.js Features

I want to get better at using advanced Node.js features like middleware, error handling, and working with databases. This will help me manage server tasks more effectively and create more reliable applications. Improving these skills will help me build more robust and scalable server-side applications.

Tailwind CSS Customization

I aim to get better at customizing and extending Tailwind CSS for specific design needs. This includes creating custom themes and optimizing performance to make my user interfaces more unique and tailored.

Action I can take to improve my skills and learning (make these “SMART”):

Advanced Node.js

Goal: To deepen my knowledge and application of advanced Node.js features.

Action: I will research advanced Node.js topics through tutorials, documentation, and online communities. I will also build small projects to practice middleware, error handling, and database integration.

Timing: By the end of the fourth week.

Tailwind CSS Customization

Goal: To enhance my ability to customize and extend Tailwind CSS for unique design needs.

Action: I will study advanced Tailwind CSS features through official documentation, online tutorials, and examples. I will apply these customizations to my projects and seek feedback to refine my skills.

Timing: By the end of the second week.

LEARNING LOG 3

For the period 15th July, 2024 to 23rd July, 2024.

What have I done? (relate tasks to Learning Outcomes)

(LO3) – Problem-Solving Skills

While working on the CRUD application, I faced some challenges, especially with handling data operations and making sure the frontend and backend communicated well. I had trouble getting the update and delete functions to work correctly. I also ran into issues when trying to connect to the API and fetch data properly. To solve these problems, I asked my coordinator for help, and with their guidance, I was able to fix the issues. This experience improved my problem-solving skills and taught me the value of asking for help when needed.

(LO5) – React.js

This week, I focused on learning more about React.js by working with the `useState` and `useEffect` hooks. I used `useState` to keep track of data in my components and `useEffect` to manage things like fetching data from APIs and updating the page. I also started learning about more advanced topics like creating custom hooks and handling state in bigger applications. These skills are important for making my web applications more interactive and responsive to user actions.

(LO7) – Node.js

I focused on building a basic CRUD (Create, Read, Update, Delete) application using Node.js. I created a card system where users can add, view, update, and delete cards.

Initially, I faced difficulties in implementing the update and delete functions correctly, but after some research and trial and error, I managed to get everything working smoothly. This project helped me gain practical experience in building server-side applications with Node.js.

(LO6) – API Integration

This week, I worked on integrating APIs into my project, using MySQL to manage the database. I faced some challenges while trying to connect the API and retrieve data correctly. Through persistence and troubleshooting, I learned how to efficiently fetch data from the API and display it in my application. This experience strengthened my understanding of how to connect frontend and backend systems effectively.

What I did well (refer to skills used)

Mastering React Hooks

I effectively used React's useState and useEffect hooks to manage component state and handle side effects, such as fetching data and updating the DOM. My ability to apply these hooks correctly enhanced the functionality and responsiveness of my React components, making the user interface more dynamic.

API Integration with Node.js and MySQL

I successfully integrated APIs with Node.js, using MySQL as the backend database. The development of a CRUD application demonstrated my capability to manage data operations and ensure seamless communication between different parts of the application, an essential skill in full-stack development.

Effective Problem-Solving

When I faced difficulties in implementing CRUD operations, particularly with the update and delete features, I sought help from my coordinator. Their assistance helped me troubleshoot the issues effectively, which not only solved the problem but also enhanced my problem-solving skills and reinforced the importance of collaboration.

What I could improve on (i.e. skills I want to improve)

Advanced React Concepts

I want to improve my understanding of advanced React concepts, such as context API, performance optimization, and managing state in larger applications. This will help me build more efficient and maintainable React applications.

Optimizing MySQL Queries

I aim to enhance my ability to write optimized MySQL queries for better performance, especially when dealing with large datasets. Understanding query optimization techniques will allow me to build more efficient and scalable database-driven applications.

Action I can take to improve my skills and learning (make these “SMART”):

Advanced React Concepts

Goal: To deepen my knowledge of advanced React concepts.

Action: I will study advanced topics like the context API and performance optimization through online courses and tutorials. I will also apply these concepts in my projects to solidify my understanding.

Timing: By the end of the fourth week.

Optimizing MySQL Queries

Goal: To improve my ability to write optimized MySQL queries for improved performance.

Action: I will research query optimization techniques, such as indexing and query refactoring, and apply them to my existing projects. I will also test different approaches to understand their impact on performance.

Timing: By the end of the fourth week.

LEARNING LOG 4

For the period 24th July, 2024 to 1st August,2024

What have I done? (relate tasks to Learning Outcomes)

(LO2) – Time Management

This week, I worked on improving my time management by maintaining a structured schedule for my tasks. I ensured punctuality in attending office and completing assigned work on time. By setting clear priorities, I was able to finish tasks efficiently without unnecessary delays. Proper planning allowed me to balance multiple responsibilities effectively and meet my deadlines without stress.

(LO7) – Node.js

This week, I implemented a basic login and registration system in Node.js. I connected my backend to a MySQL database and ensured secure storage of user credentials. I also worked on improving the structure of my backend code by following best practices, such as separating routes and controllers for better maintainability. Additionally, I integrated my previously built CRUD operations with the frontend, allowing seamless data flow between the client and server.

(LO4) – Research Skills

I focused on researching authentication methods, including login and registration functionality. I explored best practices for handling user authentication securely, including hashing passwords, using authentication tokens, and implementing session management. I used online documentation, tutorials to understand how authentication works in real-world applications. This research helped me apply authentication principles correctly while working on my Node.js backend.

(LO5) – React.js

I worked on connecting the frontend with the backend for user authentication. I integrated login and registration features into my React application, ensuring that users could sign up and log in successfully. I handled authentication responses from the backend and displayed appropriate messages based on success or failure. This practice strengthened my knowledge of React.js and how it interacts with backend APIs.

What I did well (refer to skills used)

Time Management

I stayed on track with my tasks, meeting deadlines and maintaining productivity without feeling stressed.

Node.js Development

I implemented a working authentication system and integrated it with MySQL, demonstrating my ability to build a secure backend.

React.js

I connected the frontend with the backend in React.js, making user authentication fully functional in my project.

What I could improve on (i.e. skills I want to improve)

Advanced Security Measures

While I implemented basic authentication, I want to improve by learning more about JWT (JSON Web Tokens), and multi-factor authentication.

Error Handling

I want to refine how errors are handled in both the frontend and backend, ensuring users receive clear feedback and that unexpected issues are managed properly.

Action I can take to improve my skills and learning (make these “SMART”):

Advanced Security in Authentication

Goal: To implement stronger authentication and security measures in my Node.js backend.

Action: I will research and implement JWT-based authentication, password hashing with bcrypt, and session management. I will also test different security approaches in a small project.

Timing: By the end of the fifth week.

Improving Error Handling

Goal: To enhance my ability to handle errors gracefully in both frontend and backend.

Action: I will study best practices for error handling, implement structured error messages in my Node.js API, and improve error handling in my React components.

Timing: By the end of the fifth week.

LEARNING LOG 5

For the period 5th July, 2024 to 13th August, 2024

What have I done? (relate tasks to Learning Outcomes)

(LO7) – Node.js

This week, I focused on middleware, cookies, and authentication in Node.js. I learned how middleware functions help process requests before they reach the main logic, making applications more organized and secure. I implemented authentication middleware to verify user sessions before allowing access to protected routes. Additionally, I worked with cookies and session management to store user login details securely. These concepts were essential for maintaining authentication across different pages, ensuring that users stay logged in until they choose to log out.

(LO3) – Problem-Solving Skills

While implementing authentication, I faced several challenges, such as handling expired tokens and maintaining user sessions across page reloads. Debugging these issues required a deeper understanding of JWT tokens, cookies, and session storage. I initially struggled to ensure that users remained logged in even after refreshing the page. To overcome this, I researched solutions online and consulted my coordinator for guidance. With proper debugging and testing, I successfully resolved these problems, improving both my problem-solving and debugging skills.

(LO5) – React.js

In React.js, I improved the login and logout functionality by integrating authentication mechanisms with the backend. I worked on storing authentication tokens in cookies and managing user sessions within the React application. I also ensured that authenticated users could access protected routes while preventing unauthorized users from viewing sensitive content. Implementing logout functionality properly helped me understand how to clear cookies and reset user sessions effectively.

What I did well (refer to skills used)

Implemented Authentication Middleware

I stayed on track with my tasks, meeting deadlines and maintaining productivity without feeling stressed.

Used Cookies for Session Management

I implemented a working authentication system and integrated it with MySQL, demonstrating my ability to build a secure backend.

Enhanced React Authentication

I connected the frontend with the backend in React.js, making user authentication fully functional in my project.

What I could improve on (i.e. skills I want to improve)

Advanced Middleware Usage

While I implemented basic authentication middleware, I want to explore more advanced middleware concepts, such as error-handling middleware and third-party authentication libraries.

Secure Cookie Storage

I want to learn more about securing cookies using httpOnly and SameSite attributes to prevent vulnerabilities like CSRF attacks.

Action I can take to improve my skills and learning (make these “SMART”):

Advanced Middleware Implementation

Goal: To learn more about middleware by adding error-handling and third-party authentication.

Action: I will research and use OAuth authentication with Passport.js.
Timing: Complete by the end of the sixth week.

Timing: Complete by the end of the seventh week.

Improving Cookie Security

Goal: To make authentication more secure by properly setting up cookies.

Action: I will learn about secure cookie storage and apply httpOnly and SameSite settings.

Timing: Complete by the end of the seventh week.

LEARNING LOG 6

For the period 15th July, 2024 to 25th August, 2024

What have I done? (relate tasks to Learning Outcomes)

(LO7) – Node.js

I spent time learning more about handling file uploads in Node.js, specifically using Multer. I implemented a feature where users can upload photos, which are then saved to the database. This required me to create an API endpoint for uploading files and saving metadata to the database. I also tested the functionality using Postman to ensure everything worked as expected.

(LO6) – API Integration

I focused on integrating APIs with my frontend application. I connected React.js with Node.js APIs to fetch, display, and manipulate data. I used tools like Axios to handle HTTP requests, ensuring smooth data flow between the frontend and backend. Additionally, the office provided us with Postman to test the APIs. I learned how to use Postman effectively to send different types of requests to my API and inspect the responses. This practice helped me ensure that my API endpoints were functioning correctly before connecting them to the frontend.

(LO5) – React.js

This week, I enhanced my React.js skills by integrating APIs into my frontend application. I learned how to fetch data from the backend and display it in a user-friendly way. Using state management, I ensured that the data was properly updated when the component rendered. This experience helped me further refine my skills in building interactive user interfaces that can react to user input and backend changes.

What I did well (refer to skills used)

React.js and API Integration

I successfully integrated React.js with Node.js APIs, which allowed me to fetch and display data seamlessly. This integration helped improve my understanding of how the frontend and backend communicate, especially in full-stack development.

File Upload with Multer

I implemented photo upload functionality using Multer, allowing users to upload images that were stored in the database. The task of setting up file handling and managing storage paths was a good challenge, and I successfully overcame it by following best practices.

Testing with Postman

I learned how to use Postman for testing my API endpoints. The office provided us with Postman to test the APIs, and I became proficient at sending requests (GET, POST, PUT, DELETE) to ensure that each endpoint was functioning as expected. This helped me identify and fix any issues early, making my development process more efficient.

What I could improve on (i.e. skills I want to improve)

Error Handling with File Uploads

While I successfully implemented the upload functionality, I want to improve my error-handling skills, especially when dealing with file uploads. This includes adding checks for file size, type, and ensuring a smooth user experience in case of errors.

API Security

I need to learn more about securing my APIs, especially when handling file uploads.

Ensuring proper authentication and validation for user requests will help prevent malicious uploads and protect the system.

Action I can take to improve my skills and learning (make these “SMART”):

Error Handling with File Uploads

Goal: Improve error handling in file upload functionality.

Action: I will research and implement proper error checks for file uploads (e.g., size limit, type check, and handling failed uploads).

Timing: Complete by the end of the seventh week.

API Security

Goal: Learn and implement security measures for APIs handling file uploads.

Action: I will study API security best practices and implement security features like user authentication, input validation, and anti-malware checks.

Timing: Complete by the end of the seventh week.